TEXAS A&M UNIVERSITY

Multi-Class Tumor Classification from RNA-Seq Gene Expression

TEAM

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COURSE · [STAT 654] **DATE** · [22 April 2026]

Why cancer classification matters

Different cancers need different treatments — getting the type right is critical.

- **Targeted Treatment:** Cancer care is tissue-specific; a lung cancer protocol is ineffective for kidney cancer .
- **Pathology Assistance:** Machine learning acts as a powerful "second opinion" to help pathologists.
- **Analyzing Molecular Fingerprints:** Every tumor type leaves a unique gene expression signature that mostly our models might identify and classify.
- **Handling Complexity:** ML manages the massive scale of 20,000+ gene features per patient

Project goals and objectives

What we set out to do in this project.

MAIN GOAL Build an accurate ML classifier that distinguishes 5 cancer types from gene expression data.

Specific objectives

1. Refine the feature space by eliminating low-variance and constant genes.
2. Characterize unique molecular signatures for five cancer types using dimensionality reduction
3. Compare multiple machine learning algorithms
4. Minimize the generalization gap through systematic hyperparameter tuning
5. Evaluate diagnostic reliability across imbalanced classes using confusion matrix.

Dataset overview

TCGA PAN-Cancer dataset from the UCI Machine Learning Repository.

801
Patient samples

20,531
Gene expression features

5
Cancer types

0
Missing values

The five cancer types

BRCA	Breast cancer
KIRC	Kidney renal clear cell
LUAD	Lung adenocarcinoma
PRAD	Prostate adenocarcinoma
COAD	Colon adenocarcinoma

Class distribution

How the 801 samples are split across the 5 cancer types.

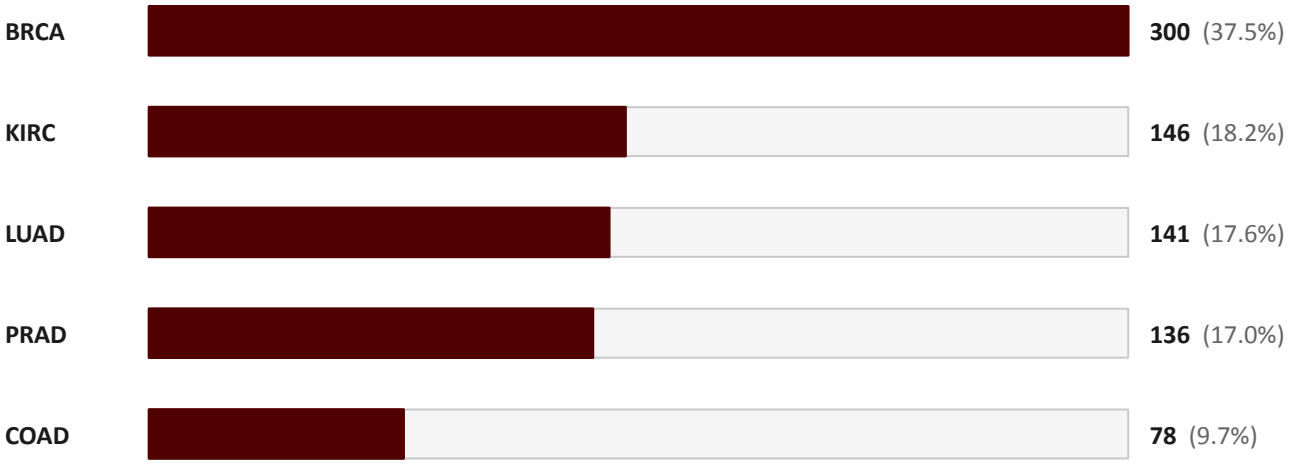


Figure 1. Sample count per cancer type (n = 801).

Key observations

- BRCA is largest (~37%)
- COAD is smallest (~10%)
- ~3.85× imbalance ratio
- Moderate, not severe
- Use stratified split

Data quality and preprocessing

Cleaning the raw data before modeling.

Initial quality checks

- ✔ 0 missing values
- ✔ 0 duplicate rows
- ✔ Values already $\log_2$-normalized
- ✔ Expression range: 0.00 – 20.78

Cleaning steps

- ➢ Removed 267 constant genes (no variation)
- ➢ Removed 43 low-variance genes ($\sigma^2 < 0.001$)

RESULT **801 samples × 20,221 informative genes**

Data characteristics and challenges

Three issues we had to address before modeling.

1 High dimensionality

CHALLENGE

20,221 features vs. only 801 samples.

OUR SOLUTION

Feature selection and PCA to reduce dimensions.

2 Class imbalance

CHALLENGE

BRCA (300) vs. COAD (78) ~ 3.85× ratio.

OUR SOLUTION

Stratified train/test splits preserve class ratios.

3 Gene correlation

CHALLENGE

Many genes are co-expressed in pathways — redundant.

OUR SOLUTION

PCA captures shared variation cleanly.

Gene expression patterns

Visual evidence that cancer types separate by their expression signatures.

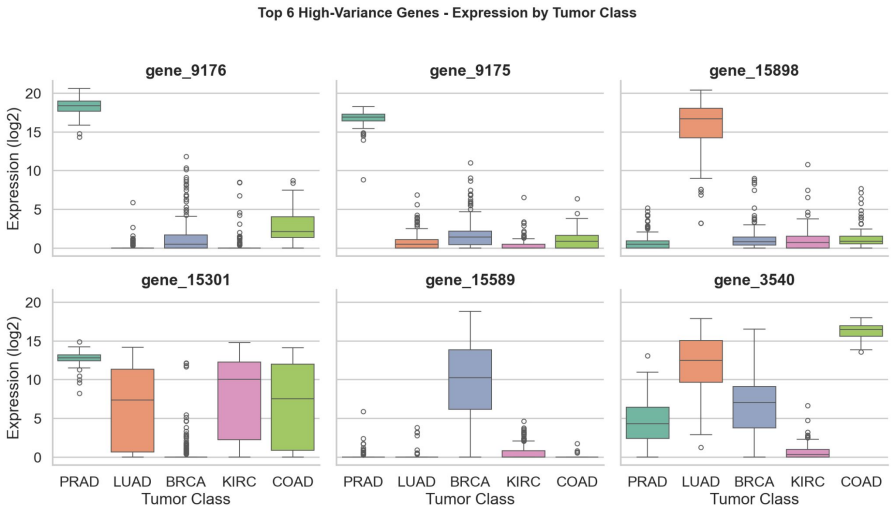


Figure 2a. Top-6 high-variance genes by tumor class.

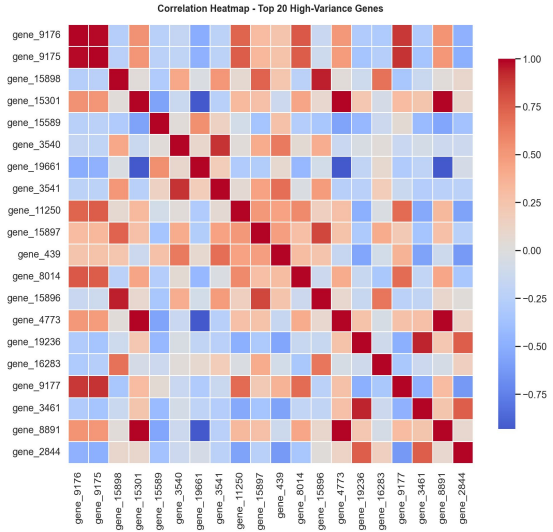


Figure 2b. Correlation heatmap — top-20 variance genes.

Observations

- Clear separation between distinct tumor types
- Each cancer type shows a unique molecular signature
- Strong correlation across many gene clusters
- Patterns suggest underlying gene networks and pathways

Statistical feature selection — ANOVA

Ranking genes by how well they separate the 5 cancer classes.

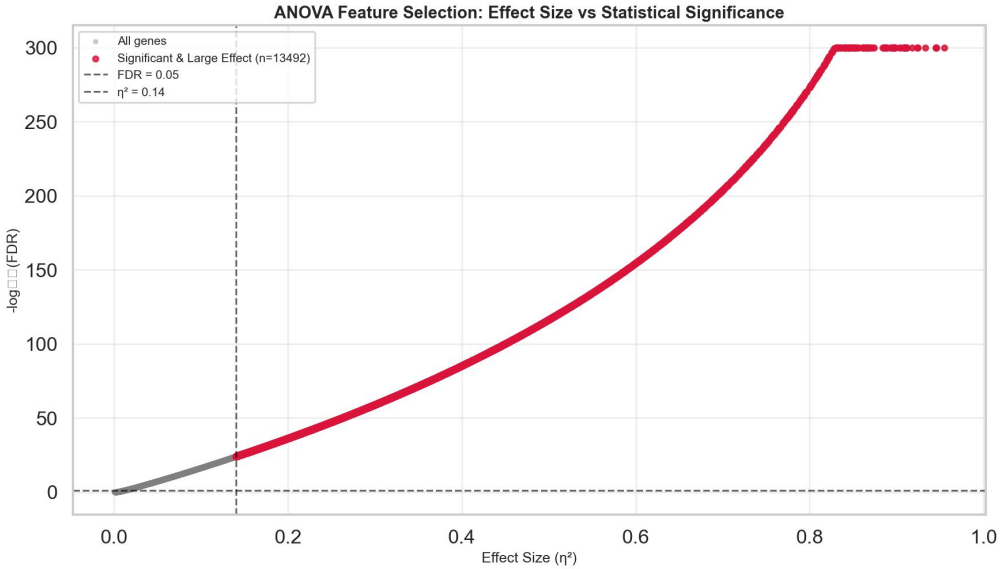


Figure 3. ANOVA feature selection — effect size vs. statistical significance.

Data overview

Total genes analyzed

20,221

Significant genes (FDR < 0.05)

19,564 (96.8%)

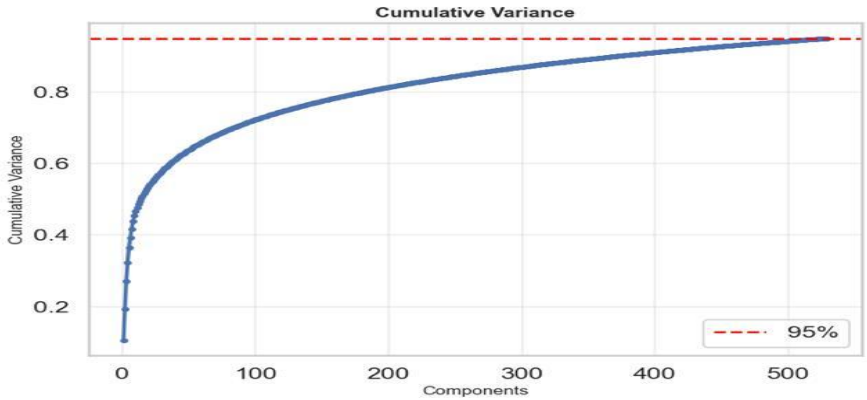
Large effect size ($\eta^2 \geq 0.14$)

13,492 (66.7%)

High proportion of genes show significant differential expression with substantial effect sizes across tumor classes.

Principal Component Analysis (PCA)

How much variance is captured by each principal component.



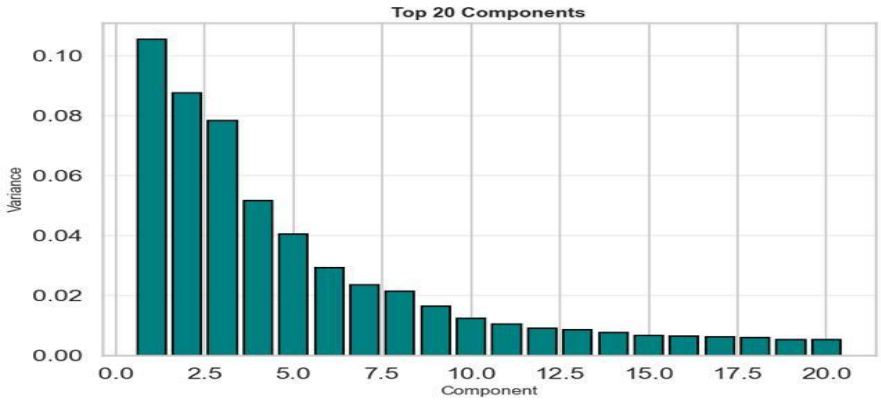


Figure 4. Cumulative variance (left) and top-20 component variance (right).

- Primary components PC1 and PC2 account for 20.4% of total variance.
- PC1 explains 11.2%, while PC2 captures 9.2% of data variability.
- Dimensionality reduction is modest — ~500 PCs are required to reach 95% cumulative variance.

Principal Component Analysis (PCA)

2D and 3D projections show how well the classes separate.

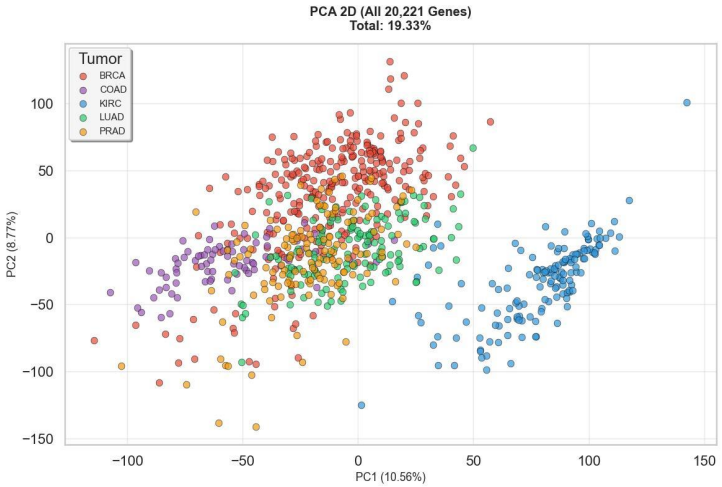


Figure 5a. PCA 2D projection.

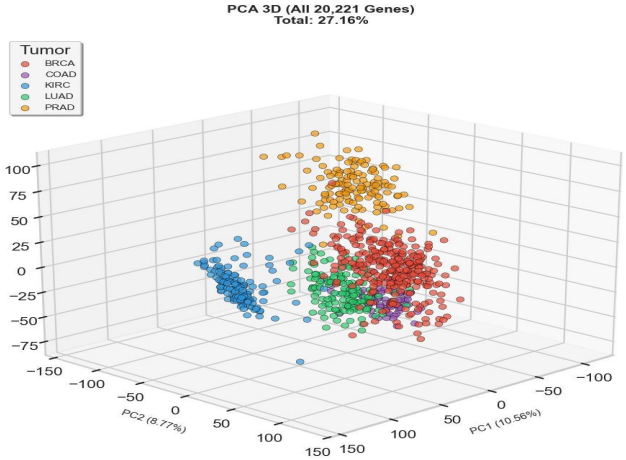


Figure 5b. PCA 3D projection.

Key takeaways

- KIRC samples show distinct clustering — well-separated with only PC1 & PC2.
- Other classes (e.g., PRAD) separate better as more dimensions (3D) are added.

ML implementation

Our modeling approach and why we compared multiple algorithms.

Objectives

- Train multiple classification models
- Compare model performance with cross-validation
- Evaluate models on the test set
- Select the best-performing model
- Perform hyperparameter tuning
- Identify the final best-performing model

Why use different models

- Compare performance across algorithms
- Different models capture different patterns
- Avoid bias from a single model choice
- Select the best-generalizing model

ML implementation — algorithms compared

Seven classification algorithms benchmarked side-by-side.

Algorithms

- Gradient Boost
- Decision Tree
- Random Forest
- Logistic Regression
- SVM
- Naive Bayes
- K-Nearest Neighbors

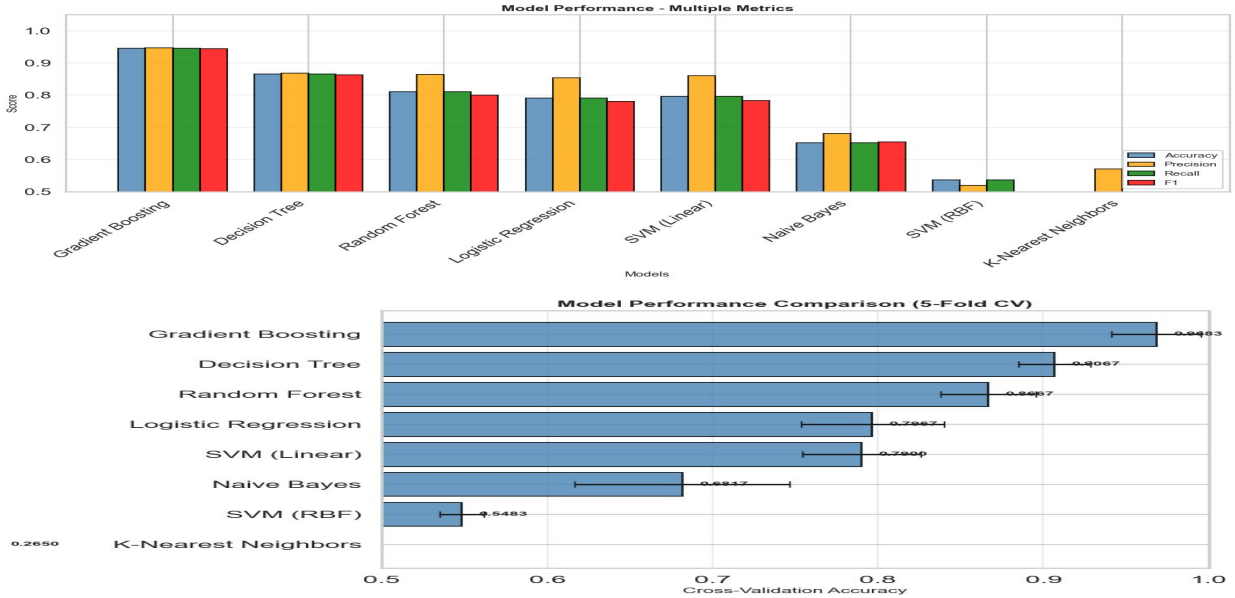


Figure 6. Model performance — multi-metric (top) and 5-fold CV accuracy (bottom).

Hyperparameter tuning and cross-validation

Optimizing the best model with GridSearchCV.

Hyperparameter optimization

Method: GridSearchCV

Parameters tested

- n_estimators: [50, 100, 200]
- learning_rate: [0.01, 0.1, 0.2]
- max_depth: [3, 5, 7]
- min_samples_split: [2, 5]

Best parameters found

- n_estimators: 100
- learning_rate: 0.1
- max_depth: 5
- min_samples_split: 2

5-fold cross-validation results

96.83%	94.53%	2.3%
CV accuracy	Test accuracy	CV – Test gap

Performance improvement

~88%	94.53%	+6.53%
Before tuning	After tuning	Improvement

Best performing model — Gradient Boosting

Confusion matrix and why this model won.

Why this model won

- Highest CV accuracy: 96.83%
- Best test accuracy: 94.53%
- Outperformed 7 other algorithms

Optimized hyperparameters

- n_estimators: 100
- learning_rate: 0.1
- max_depth: 5
- min_samples_split: 2

Model characteristics

- Type: Ensemble (sequential trees)
- Training time: ~3–5 seconds · Inference: <1 ms per sample · Model size: compact

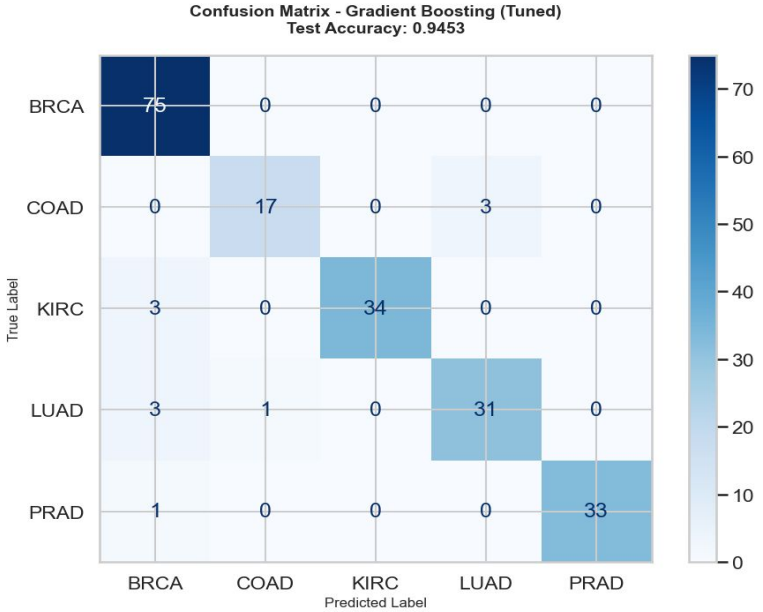


Figure 7. Confusion matrix — Gradient Boosting (tuned), test accuracy 94.53%.

Future scope

Where this project could go next.

1	**External validation**	Test model on independent datasets from other institutions.
2	**Expand cancer types**	Scale from 5 to 10–20 different tumor types.
3	**Deep learning exploration**	Implement neural networks for improved accuracy.
4	**Multi-omics integration**	Combine RNA, DNA, and protein data for comprehensive profiling.
5	**Clinical deployment**	Develop a real-time diagnostic tool for hospital use.

Thank You

Questions & Discussion